PUSHING A FINITE TYPE IDEAL ALONG AN ETALE RING MAP INTO A PRODUCT INVOLVING AN IDEAL OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0813 (sources/stacks-flat/pinned-original-source.tex line 8876).

Allowed standard prerequisites (editorial).

- An etale ring map is standard etale after localising at a suitable element. Algebra, Proposition labelled algebra-proposition-etale-locally-standard (Stacks tag 00UE), invoked at proof lines 8888-8889. Supplies the reduction to B = C_h with C = A[x]/(f) for a monic f of degree d, which is what makes the exponent dm computable.
- The fitting ideal trick relating an ideal to a fittable presentation. Same chapter, Lemma labelled lemma-trick-fitting-ideal (source line 8849), invoked at proof line 8896. Supplies the recipe I = Fit_0(C/(h_1, ..., h_{r+1})) used to produce the ideal of the base; the verification that it has the required vanishing locus is inside the counted span.

Source reference. Original proposition 00UE: sources/stacks-chow/tags.txt, source lines 1108–1108. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8889 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0812: sources/stacks-flat/pinned-original-source.tex, source lines 8849–8849. Same-chapter pointer into the author environment lemma at source line 8849; invoked at proof line 8896 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $A \to B$ be an étale ring map. Let $a \in A$ be a nonzerodivisor. Let $J \subset B$ be a finite type ideal with $V(J) \subset V(aB)$. For every $\mathfrak{q} \subset B$ there exists a finite type ideal $I \subset A$ with $V(I) \subset V(a)$ and $g \in B$, $g \notin \mathfrak{q}$ such that $IB_g = JJ'$ for some finite type ideal $J' \subset B_g$.*

Proof. We may replace B by a principal localization at an element $g \in B$, $g \notin \mathfrak{q}$. Thus we may assume that B is standard étale, see Algebra, Proposition 00UE. Thus we may assume B is a localization of $C = A[x]/(f)$ for some monic $f \in A[x]$ of some degree d. Say $B = C_h$ for some $h \in C$. Choose elements $h_1, \ldots, h_n \in C$ which generate J over B. The condition $V(J) \subset V(aB)$ signifies that $a^m = \sum b_i h_i$ in B for some large m. Set $h_{n+1} = a^m$. As in Lemma 0812 we take $I = \mathrm{Fit}_0(C/(h_1, \ldots, h_{r+1}))$. Since the module $C/(h_1, \ldots, h_{r+1})$ is annihilated by a^m we see that $a^{dm} \in I$ which implies that $V(I) \subset V(a)$. □